

Note

Lab11 Exercise

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Outline

- 00_TESTBED
- 01_RTL
- 02_SYN
- 03_GATE
- 04_MEM
- 05_APR
- 06_POST



00_TESTBED

- Finish the pattern in 00_TESTBED

```
`ifdef RTL
    `define CYCLE_TIME 20.0
`elsif GATE
    `define CYCLE_TIME 20.0
`elsif POST
    `define CYCLE_TIME 20.0
`endif
```

In this lab, you should submit two cycle time.

One of them is for 01_RTL & 03_GATE (determined in 02_SYN)

The other is for 06_POST (determined in 05_APR)

The two cycle time can be different or the same

- Define the path of all memory.v in filelist.f

```
1 TESTBED.v
2 ../04_MEM/SUMA180_4096X8X1BM4.v
3 ../04_MEM/SUMA180_2048X16X1BM1.v
4 ../04_MEM/SUMA180_1024X32X1BM2.v
```

Three types of SRAMs generated by TA are defined in filelist.f

If you have your own SRAMs, you should add and modify them in this file.



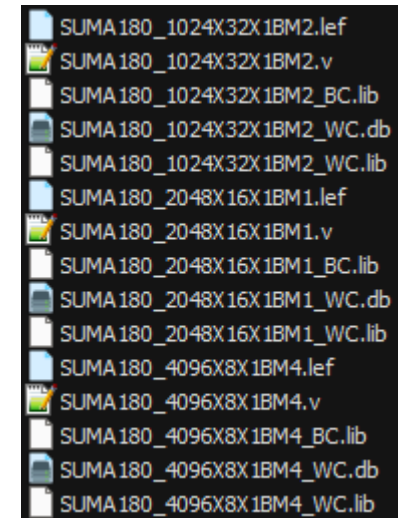
01_RTL

- Finish the RTL code: GTE.v
- Prepare all memory.v in 04_MEM

All necessary files related to SRAMs generated by TA are in 04_MEM →

If you have your own SRAMs, you should add and modify them in this file.

- Execute **./01_run_rtl**
 - Functionality correctness
 - Latency is limited to 5000 cycles
 - No error message about SRAM



The memory files used in this lab

1. memory.v
2. memory_WC.db
3. memory_WC.lib
4. memory_BC.lib
5. memory.lef



02_SYN (1/2)

- Modify target_library and link_library in .synopsys_dc.setup

```
set target_library " fsa0m_a_generic_core_sslp62v125c.db \  
fsa0m_a_generic_core_fflp98vm40c.db \  
fsa0m_a_t33_generic_io_sslp62v125c.db \  
fsa0m_a_t33_generic_io_fflp98vm40c.db \  
SUMA180_1024X32X1BM2_WC.db \  
SUMA180_2048X16X1BM1_WC.db \  
SUMA180_4096X8X1BM4_WC.db "  
  
set link_library " * $target_library dw foundation.sldb standard.sldb \  
SUMA180_1024X32X1BM2_WC.db \  
SUMA180_2048X16X1BM1_WC.db \  
SUMA180_4096X8X1BM4_WC.db"
```

Three types of SRAMs generated by TA are defined in this files

If you have your own SRAMs, you should add and modify them in this file.



02_SYN (2/2)

- Modify the synthesis cycle time in syn.tcl
 - Maximum allowed cycle time is 20.0 ns
 - Precision of clock is 0.1
- Execute **./01_run_dc_shell**
 - No Latch & Mismatch width of memory
 - Total cell area should not larger than 5,000,000 μm^2
 - Macro/Black Box cannot be smaller than 2,100,000 μm^2
 - Synthesis time should be less than 4 hours



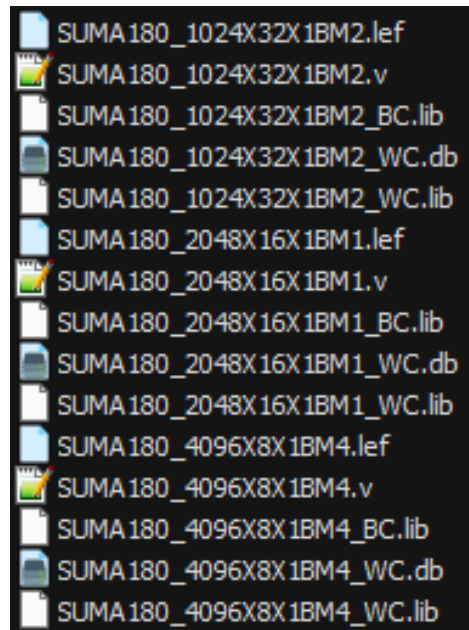
03_GATE

- Execute **./01_run_gate**
 - Functionality correctness
 - Latency is limited to **5000** cycles
 - No error message about SRAM



04_MEM (1/2)

- Prepare all_your_memory.v / .db / .lef / .lib in 04_MEM



The memory files used in this lab

1. memory.v
2. memory_WC.db
3. memory_WC.lib
4. memory_BC.lib
5. memory.lef



04_MEM (2/2)

- You should modify the lef files in 04_MEM like below:

```

NAMECASESENSITIVE ON ;
MACRO MEM_IMAGE
CLASS BLOCK ;
FOREIGN MEM_IMAGE 0.000 0.000 ;
ORIGIN 0.000 0.000 ;
SIZE 994.480 BY 176.400 ;
SYMMETRY x y r90 ;
SITE core ;
PIN VCC
  DIRECTION INOUT ;
  USE POWER ;
  SHAPE ABUTMENT ;
PORT
  LAYER ME4 ;
  RECT 993.360 164.980 994.480 168.220 ;
  LAYER ME3 ;
  RECT 993.360 164.980 994.480 168.220 ;
  LAYER ME2 ;
  RECT 993.360 164.980 994.480 168.220 ;
  LAYER ME1 ;
  RECT 993.360 164.980 994.480 168.220 ;
END
  LAYER VI1 ;
  RECT 0.000 0.140 994.480 176.400 ;
  LAYER VI2 ;
  RECT 0.000 0.140 994.480 176.400 ;
  LAYER VI3 ;
  RECT 0.000 0.140 994.480 176.400 ;

```

Raw lef files

- Modify all the Memory.lef file
 - Change “core” to “core_5040”
 - Change “ME1” to “metal1” and so on
 - Change “VI1” to “via”
 - Change “VI2” to “via2” and so on



```

NAMECASESENSITIVE ON ;
MACRO SUMA180_1024X32X1BM2
CLASS BLOCK ;
FOREIGN SUMA180_1024X32X1BM2 0.000 0.000 ;
ORIGIN 0.000 0.000 ;
SIZE 1005.020 BY 294.000 ;
SYMMETRY x y r90 ;
SITE core 5040 ;
PIN VCC
  DIRECTION INOUT ;
  USE POWER ;
  SHAPE ABUTMENT ;
PORT
  LAYER metal4 ;
  RECT 1003.900 282.580 1005.020 285.820 ;
  LAYER metal3 ;
  RECT 1003.900 282.580 1005.020 285.820 ;
  LAYER metal2 ;
  RECT 1003.900 282.580 1005.020 285.820 ;
  LAYER metal1 ;
  RECT 1003.900 282.580 1005.020 285.820 ;
END
  LAYER via ;
  RECT 0.000 0.140 1005.020 294.000 ;
  LAYER via2 ;
  RECT 0.000 0.140 1005.020 294.000 ;
  LAYER via3 ;
  RECT 0.000 0.140 1005.020 294.000 ;

```

Modified lef files

Three types of SRAMs generated by TA have been modified
 If you have your own SRAMs, you should modify them



05_APR (1/2)

- Put all modified Memory.lef in **05_APR/LEF**
- Put all Memory_WC.lib & Memory_BC.lib in **05_APR/LIB**
- Follow APR steps in **APR_flow_2025F.pdf**
 - You can execute `./01_setenv` to set OA_HOME

```
05_APR]$ ./01_setenv  
#!/bin/csh  
setenv OA_HOME /RAID2/cad/cadence/INNOVUS/INNOVUS_20.15.000/share/oa
```

- You can `source apr_setting.cmd` after you activate Innovus

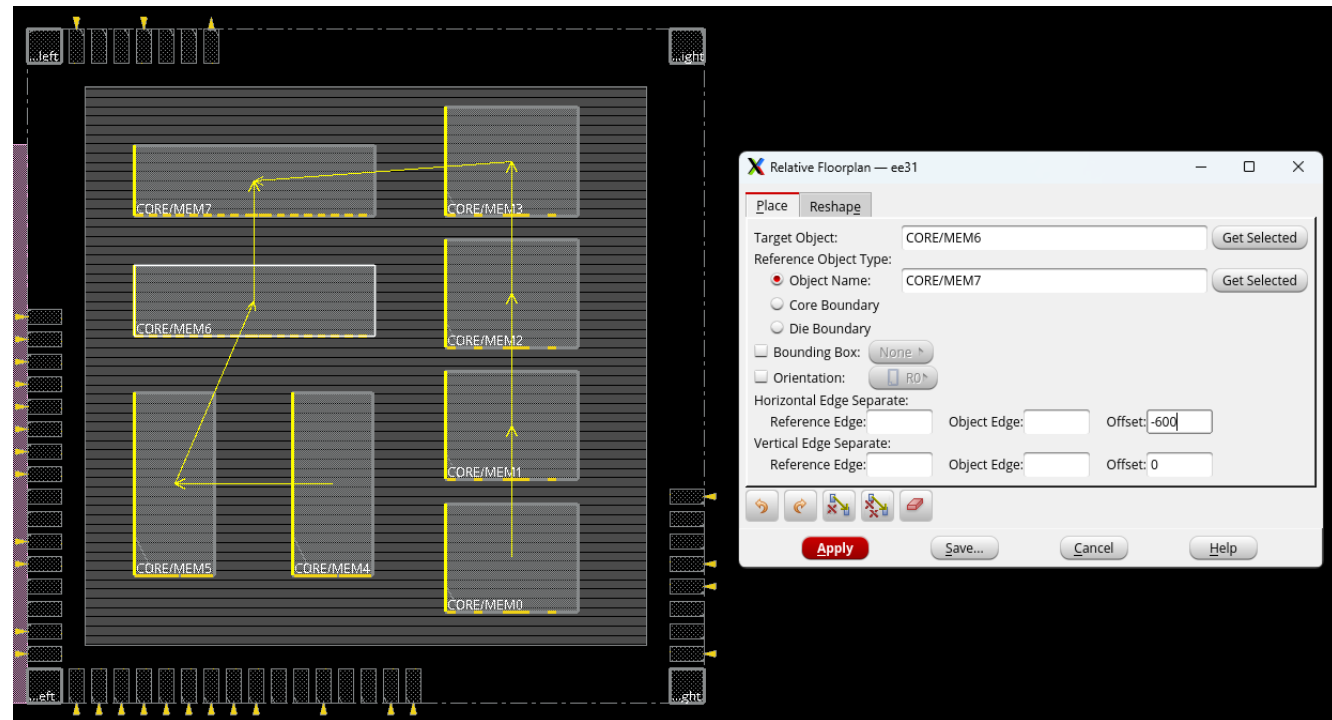
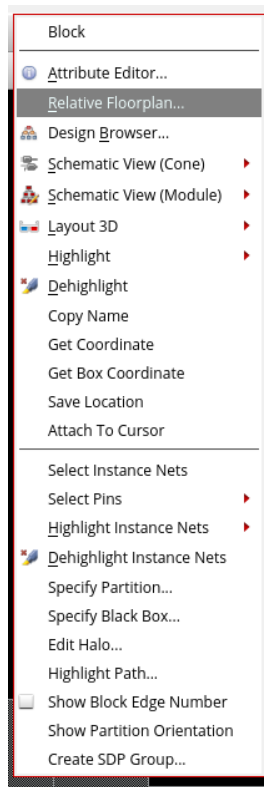
```
[INFO] Loading Pegasus 21.21 fill procedures  
innovus 1> source ./cmd/apr_setting.cmd
```

```
#####  
##### APR UMC018 setting #####  
#####  
set init_design_uniquify 1  
setDesignMode -process 180  
suppressMessage TECHLIB 1318  
suppressMessage ENCEXT-2799
```



05_APR (2/2)

- Floorplan – Relative floorplan



06_POST

- Execute ./01_run_post
 - Functionality correctness
 - Latency is limited to 5000 cycles
 - No error message about SRAM
 - Latency of 06_POST should be the same as that of 03_GATE



Thanks

